



Advanced Tools & Automation

A Complete Textbook on Macros, VBA, Dashboards & More

Chapter 1

Introduction to Macros

Welcome to the world of automation in Microsoft Excel! This chapter introduces you to one of the most powerful yet often overlooked features of Excel — Macros. By the end of this chapter, you will understand what macros are, why they are useful, and how to record and run them without writing a single line of code.

1.1 What is a Macro?

Imagine you perform the same sequence of steps in Excel every morning — you open a file, apply a specific formatting style, sort a column, add a total row, and save. If this takes you 10 minutes each day, that adds up to over 40 hours in a year — spent on completely repetitive work.

A Macro eliminates this repetition entirely by recording your actions once and replaying them instantly, every time you need them.

1.2 Why Use Macros?

Macros offer four key advantages that make them valuable in any professional environment:

1.3 The Developer Tab — Your Macro Control Panel

Before you can record or manage macros, you need to enable the Developer tab in Excel. By default, Excel hides this tab because it is intended for more advanced users.

How to Enable the Developer Tab

1. Open Excel. Click on the File menu in the top-left corner.
2. Click Options at the bottom of the left-side menu.
3. In the Excel Options window, click Customize Ribbon on the left.
4. On the right side, you will see a list of main tabs. Look for Developer and tick the checkbox next to it.
5. Click OK to close the window.

Once enabled, the Developer tab appears in the Excel ribbon. It contains the following key tools:

1.4 Recording Your First Macro

Now it is time to record a macro. In this example, you will record a macro that formats a header row — making it bold, changing the background colour to dark blue, and the font colour to white. This is a task many professionals perform repeatedly.

Step-by-Step: Recording a Macro

6. Open a new Excel workbook. In the first row (Row 1), type these column headers: Employee ID, Name, Department, Salary, Join Date.
7. Click on cell A1 to position your cursor at the beginning of the header row.
8. Go to the Developer tab and click Record Macro.
9. In the Record Macro dialog box, fill in the following:
10. Click OK. Excel is now recording every action you take.
11. Select the entire Row 1 by clicking the row number '1' on the left side.
12. From the Home tab, apply Bold (Ctrl + B).
13. Go to Fill Color (the paint bucket icon) and choose a dark blue colour.
14. Go to Font Color and choose White.
15. Go to the Developer tab and click Stop Recording.

1.5 Running a Macro

Once recorded, you can run your macro in three ways:

Method 1 — Using the Macros List

16. Go to the Developer tab and click Macros.
17. In the Macro dialog box, select your macro name from the list.
18. Click Run. Excel instantly performs all the recorded steps.

Method 2 — Using a Keyboard Shortcut

If you assigned a shortcut key (e.g., Ctrl + Shift + H) when recording, simply press those keys at any time to run the macro immediately.

Method 3 — Assigning the Macro to a Button

19. Go to the Developer tab. In the Controls group, click Insert.
20. Under Form Controls, click the Button icon (first option — looks like a rectangle).
21. Draw the button on your spreadsheet by clicking and dragging.
22. When the Assign Macro dialog appears, select your macro and click OK.
23. Right-click the button and choose Edit Text to rename it (e.g., 'Format Header').
24. Click anywhere outside the button. Now simply click the button to run the macro.

1.6 Saving a File with Macros

When you try to save a workbook that contains macros, Excel will prompt you with an important message. You must save the file in the correct format — otherwise the macro will be lost.

Chapter 2

Absolute vs. Relative Referencing in Macros

When you record a macro, Excel needs to know whether the actions you perform are tied to specific cells (like always cell A1) or whether they should adapt to wherever your cursor is at the time of running. This is controlled by a setting called referencing mode, and getting it right is the key to writing macros that work correctly in all situations.

2.1 Understanding the Two Modes

2.2 Absolute Referencing — How It Works

By default, when you record a macro without activating relative referencing, Excel records every cell address absolutely. This means the macro remembers the exact cell you clicked on.

Example: Formatting a Specific Cell

Suppose you record a macro where you click on cell B2 and type 'Approved'. When you run this macro in the future, it will always go to cell B2 and type 'Approved' — regardless of where your cursor is.

2.3 Relative Referencing — How It Works

To record a macro using relative referencing, you must click the Use Relative References button on the Developer tab before you start recording. The button will appear highlighted when active.

Example: Filling in Three Cells Below the Cursor

Suppose you want a macro that, from wherever the cursor is, fills the next three cells down with the values 'Pending', 'In Progress', and 'Completed'. With relative referencing, the macro works correctly wherever you place your cursor first.

2.4 When to Use Which Mode

Choosing the correct mode depends entirely on the task. Here is a simple guide:

2.5 Mixing Both Modes in One Macro

Advanced users often mix both modes within a single macro recording — switching relative referencing on and off during the recording to achieve the best of both worlds.

Example: Monthly Report Setup Macro

Consider a macro that does all of the following in one recording:

- Goes to cell A1 and types 'Monthly Sales Report' (Absolute — always in A1).
- Moves down 2 rows from wherever A1 is and types today's date (Relative — relative to A1).
- Navigates to cell H50 and inserts a SUM formula (Absolute — always in H50).

To achieve this, the user toggles relative referencing on/off between these steps during the recording.



Chapter 3

Basic VBA Understanding (Optional Introduction)

Every macro you record in Excel is secretly converted into code — a programming language called VBA (Visual Basic for Applications). You do not need to be a programmer to use macros, but understanding the basics of VBA will make you significantly more capable. This chapter provides a beginner-friendly introduction to VBA without requiring any prior coding experience.

3.1 What is VBA?

3.2 Opening the VBA Editor

The VBA Editor (also called the Visual Basic IDE) is where you can view and edit the code behind your macros. It opens in a separate window within Excel.

How to Open the VBA Editor

25. Go to the Developer tab.
26. Click Visual Basic. The VBA Editor window will open.

Alternatively, press Alt + F11 on your keyboard — this is the fastest way to open the VBA Editor from anywhere in Excel.

The VBA Editor has three main areas you should know:

3.3 Reading a Recorded Macro's Code

When you record a macro in Excel, a 'Module' is automatically created in the VBA Editor and the code is written there. Let us look at a real example.

After recording the 'FormatHeaderRow' macro from Chapter 1, the VBA Editor would show code similar to this:

Do not worry about understanding every line immediately. Let us break down the most important parts:

3.4 Key VBA Concepts for Beginners

Even without writing code, understanding these concepts helps you make small, useful edits to recorded macros.

Variables — Storing Information Temporarily

In this example:

- Dim declares a variable (Dim is short for 'Dimension' — an old programming term for creating a variable).
- As String means the variable holds text. As Double means it holds a decimal number.
- MsgBox displays a pop-up message box on screen — useful for showing results to the user.

If...Then Statements — Making Decisions

Reading this in plain English: 'If the value in cell B2 is greater than or equal to 100,000, then show a success message. Otherwise, show the actual sales figure.'

Loops — Repeating Actions

This macro checks every row from row 2 to row 100. If the value in Column C (stock quantity) is below 10, it highlights that cell in red. Doing this manually for 100 rows would take several minutes — the macro does it instantly.

3.5 Running and Testing VBA Code

You can run VBA code directly from the VBA Editor without going back to Excel:

27. Place your cursor anywhere inside the Sub you want to run.
28. Press F5 on your keyboard, or click the green Play button (▶) in the toolbar.
29. The macro will execute immediately.

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Chapter 4

Dashboard Creation Basics

A dashboard in Excel is a single, well-designed page that presents key information at a glance. Instead of browsing through multiple sheets or scrolling through thousands of rows, a dashboard gives decision-makers everything they need in one visual summary. This chapter covers how to plan and build a professional Excel dashboard from scratch.

4.1 What is an Excel Dashboard?

4.2 Planning Your Dashboard

Before touching Excel, you must plan your dashboard on paper. Poorly planned dashboards become cluttered and confusing. A good dashboard follows a clear purpose and structure.

Step 1 — Define the Purpose

Ask: What decisions will this dashboard support? Who will use it? What are the three to five most important things they need to see?

Step 2 — Identify the Key Metrics

Key metrics (also called KPIs — Key Performance Indicators) are the numbers that matter most. Examples include: Total Revenue, Number of Orders, Average Delivery Time, Customer Satisfaction Score.

Step 3 — Decide on the Layout

A standard dashboard layout follows a simple hierarchy:

- Top section: KPI summary cards — large numbers at a glance.
- Middle section: Charts — trends, comparisons, distributions.
- Bottom section: Data table — detailed breakdown for reference.

4.3 Setting Up the Dashboard Sheet

A professional dashboard separates data from presentation. You should have at minimum two sheets in your workbook:

Formatting the Dashboard Sheet

30. Hide gridlines: Go to View tab → uncheck Gridlines. This gives the dashboard a clean, professional look.
31. Set a background colour: Select all cells (Ctrl+A), then apply a light grey or white fill from the Home tab.
32. Remove row and column headers if preferred: Go to View tab → uncheck Headings.

4.4 Creating KPI Cards

KPI cards are simple boxes that display one important number prominently. They are the first thing a viewer looks at on a dashboard.

Method: Using Merged Cells and Formulas

33. Select a 3×2 range of cells (e.g., B2:D3). Merge them using Home → Merge & Center.
34. Type a formula that pulls data from the Data sheet: =Data!B2 or =SUM(Data!C2:C100).
35. Format the number: Apply currency, percentage, or comma-separated format from the Home tab.
36. Increase the font size to 24–36pt to make it prominent. Apply bold formatting.
37. Add a label below: In the row beneath, type the metric name (e.g., 'Total Revenue') in a smaller font.
38. Apply a coloured background to the merged cells to visually separate the KPI card.

4.5 Linking Dashboard Elements to Data

All charts and formulas on the Dashboard sheet should pull their data from the Data sheet — never entered manually. This ensures the dashboard updates automatically when new data arrives.

Formula Linking Example

If your total revenue is calculated on the Data sheet in cell D100, link it to your dashboard KPI card with:

If your KPI card needs to sum a range from the Data sheet:

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Chapter 5

Charts — Advanced Charts & Combo Charts

Charts are one of the most powerful communication tools in Excel. A well-chosen chart can convey in seconds what a table of numbers takes minutes to interpret. This chapter covers advanced charting techniques including chart customization, choosing the right chart type, and creating combination charts.

5.1 Choosing the Right Chart Type

The most common mistake in data visualization is using the wrong chart type. The chart you choose must match the story your data tells.

5.2 Building a Column Chart — Step by Step

Column charts are the most commonly used chart type in business settings. Let us build one from scratch.

Sample Data Setup

In your Data sheet, set up this sample table:

39. Select the data range (A1:B7 including headers).
40. Go to Insert tab → Charts group → Click on the Column Chart icon.
41. Select Clustered Column from the dropdown.
42. Excel inserts a basic column chart. Now let us format it professionally.

Customizing the Chart — Professional Formatting

43. Chart Title: Click on the title text 'Chart Title' and replace it with 'Monthly Sales Performance'. Increase font size to 14pt, bold.
44. Axis Labels: Right-click the vertical axis → Format Axis → change number format to Currency (₹).
45. Data Labels: Right-click any bar → Add Data Labels → Data labels appear above each bar.
46. Bar Colour: Right-click any bar → Format Data Series → Fill → choose a solid colour that matches your brand.
47. Gridlines: Click on a gridline → press Delete. Fewer gridlines = cleaner chart.
48. Legend: If there is only one data series, delete the legend — it adds clutter without value.

5.3 Creating a Combo Chart (Combination Chart)

A combo chart overlays two different chart types on the same chart — typically bars for one metric and a line for another. This is extremely useful when comparing two measurements that have different scales.

Example: Revenue vs. Profit Margin %

Set up this data table on the Data sheet:

Step-by-Step: Creating the Combo Chart

49. Select the data range A1:C7.
50. Go to Insert tab → Charts group → Click the dropdown arrow next to the column chart icon → Select All Charts.
51. In the 'Insert Chart' dialog, click Combo in the left panel.
52. You will see a configuration table at the bottom of the window. Set:
 - Revenue: Chart Type = Clustered Column, Secondary Axis = unchecked.
 - Profit Margin %: Chart Type = Line, Secondary Axis = checked.
53. Click OK. Excel creates the combo chart with bars for revenue and a line for profit margin.

Adding a Secondary Axis Manually

If you already have a chart and want to add a secondary axis later:

54. Right-click on the line or bar you want to move to a secondary axis.
55. Select Format Data Series.
56. In the Format panel, under Series Options, choose Secondary Axis.

5.4 Sparklines — Mini Charts in Cells

Inserting a Sparkline

57. Click on the cell where you want the sparkline to appear (e.g., H2).
58. Go to Insert tab → Sparklines group → Click Line, Column, or Win/Loss.
59. In the 'Data Range' field, select the cells that contain the data for this sparkline.
60. Click OK. A tiny chart appears inside the cell.



Chapter 6

Protecting Sheets & Workbooks

When you share an Excel file with colleagues or distribute it to a team, you often need to control what they can and cannot change. Excel's protection features allow you to lock specific cells, entire sheets, or the whole workbook — preventing accidental or unauthorized changes while still allowing users to interact with the parts they need.

6.1 Why Protection Matters

6.2 Protecting a Sheet

Sheet protection prevents users from editing, formatting, or deleting content on a specific worksheet.

Step 1 — Unlock the Input Cells First

By default, all cells in Excel are 'Locked'. When you protect a sheet, all locked cells become uneditable. So before protecting, you must first unlock the cells users are allowed to edit.

61. Select the cells that users should be able to edit (e.g., the data entry area B2:E50).
62. Right-click → Format Cells → go to the Protection tab.
63. Uncheck the 'Locked' checkbox → Click OK.

Step 2 — Protect the Sheet

64. Go to Review tab → Click Protect Sheet.
65. Set a password if you want to prevent others from removing the protection. (Optional but recommended.)
66. In the permissions list, specify what users ARE allowed to do (e.g., Select unlocked cells only).
67. Click OK. Re-enter the password if prompted.

6.3 Protecting the Workbook Structure

While sheet protection controls what can be edited on a sheet, workbook protection controls what can be done to the workbook itself — such as adding, deleting, moving, or renaming sheets.

68. Go to Review tab → Click Protect Workbook.
69. Check the 'Structure' option to prevent adding, deleting, or renaming sheets.
70. Enter a password (optional) and click OK.

6.4 Allowing Specific Users to Edit Ranges

For more granular control, Excel allows you to specify that only certain users (identified by their Windows username or a password) can edit specific ranges — even when the sheet is protected.

71. Go to Review tab → Click Allow Edit Ranges.

72. Click New to create a new editable range. Define the range address and set a range-specific password.
73. Repeat for each range that different user groups need to access.
74. After defining all ranges, click Protect Sheet to activate the overall protection.

6.5 Removing Protection

To remove protection from a sheet or workbook:

75. Go to Review tab → Click Unprotect Sheet (or Unprotect Workbook).
76. If a password was set, enter it when prompted.
77. The protection is immediately removed.



Chapter 7

Tips, Shortcuts & Real-World Case Study

Mastering Excel is not just about knowing features — it is about working efficiently. This chapter consolidates the most valuable keyboard shortcuts, productivity tips, and best practices used by professionals. It concludes with a full real-world case study showing how all the topics from this module come together in practice.

7.1 Essential Keyboard Shortcuts

Professional Excel users rely heavily on keyboard shortcuts to navigate and work faster. Memorising even ten shortcuts will noticeably improve your efficiency.

Navigation & Selection Shortcuts

Editing Shortcuts

Formula & Function Shortcuts

Macro-Specific Shortcuts

7.2 Professional Excel Tips

These are habits and practices that distinguish an advanced Excel user from an average one.

Tip 1 — Always Use Structured Tables

Instead of working with raw data ranges, convert data to an official Excel Table using Ctrl + T. Tables automatically expand as new rows are added, apply consistent formatting, and make formulas easier to read.

Tip 2 — Use Named Ranges for Clarity

Instead of writing =SUM(D2:D100) in formulas, name the range 'TotalSales' and write =SUM(TotalSales). This makes formulas self-documenting and reduces errors when ranges change.

Tip 3 — Freeze Panes for Large Datasets

When working with large datasets, freeze the header row and left column so they remain visible while scrolling. Go to View tab → Freeze Panes → Freeze Top Row (or Freeze First Column, or Freeze Panes for a custom combination).

Tip 4 — Use Data Validation to Control Inputs

Prevent users from entering incorrect data by setting validation rules. Select the input cells → Data tab → Data Validation. You can restrict entries to a dropdown list, a number range, or a date range.

Tip 5 — Audit Formulas with Trace Dependents / Precedents

When a formula gives a wrong result and you cannot figure out why, use Excel's auditing tools. Go to Formulas tab → Formula Auditing group:

- Trace Precedents: Shows which cells feed into the selected formula.
- Trace Dependents: Shows which cells use the selected cell's value in their formulas.
- Evaluate Formula: Walks through a formula step by step.

7.3 Real-World Case Study — Automated Monthly Sales Dashboard

This case study walks through how a regional manager at a mid-sized FMCG company used all the concepts from this module to create a fully automated monthly reporting system.

Background

The company has five sales regions. Each region submits a raw sales export every month — an unformatted Excel file with thousands of rows. The regional manager, previously spent 3–4 hours each month:

- Cleaning and formatting the raw data
- Calculating totals and averages
- Creating charts for the management presentation
- Updating the previous month's dashboard manually

After completing this Excel Advanced Tools module, he rebuilt the entire system as an automated solution.

Solution Architecture

Automation Steps Applied

The manager recorded and wrote the following macros:

- CleanRawData macro (Absolute reference): Selects fixed header cells, applies bold formatting, freezes the top row, and applies currency formatting to the Amount column. Runs in 1 second.
- GenerateMonthlyReport macro (Relative reference): Reads the selected month from the Settings sheet, filters the CleanData sheet for that month, and pastes results to a summary range. Adapts automatically to whichever month is selected.
- RefreshDashboard macro: Refreshes all pivot tables, updates chart ranges, and triggers both macros above in sequence. Assigned to a button labelled 'Update Dashboard'.

Results

Chapter 8

Final Practice — Mini Project

This chapter guides you through a comprehensive mini project that integrates every concept covered in this module. You will build a complete automated reporting workbook from scratch — using macros, VBA, a professional dashboard, advanced charts, and protection features. This project is designed to mimic a real professional scenario.

8.1 Project Overview

You are the operations analyst for a retail chain with four stores: Mumbai, Delhi, Bengaluru, and Ahmedabad. You receive a monthly raw sales export and need to produce a management-ready dashboard every month.

Project Deliverable

By the end of this project, your workbook will:

- Contain a RawData sheet where the monthly export is pasted.
- Have a Dashboard sheet with KPI cards, a bar chart, a line chart, and a combo chart.
- Include at least two macros: one with absolute referencing and one with relative referencing.
- Be protected so that only designated input cells can be edited.
- Have a one-click 'Refresh Dashboard' button that updates all elements.

8.2 Dataset for the Project

Create the following dataset on the RawData sheet. This represents 6 months of sales data for four stores:

8.3 Project Steps

Follow these steps in order. Each step references the chapter where the concept was covered.

Step 1 — Set Up the Workbook Structure (Chapter 4)

- Rename Sheet1 to 'RawData'. Create two additional sheets: 'Dashboard' and 'Settings'.
- On the Dashboard sheet: hide gridlines, hide row/column headings, set a light grey background.
- On the Settings sheet: create a dropdown list for Store selection (Mumbai, Delhi, Bengaluru, Ahmedabad, All Stores).

Step 2 — Create KPI Cards (Chapter 4)

- Create three KPI cards on the Dashboard: Total Net Revenue, Total Units Sold, Best Performing Store.
- Link each card to SUMIF or INDEX/MATCH formulas that reference the RawData sheet.
- Apply consistent colours and large font sizes to make the numbers prominent.

Step 3 — Build the Charts (Chapter 5)

- Bar chart: Net Revenue by Store for the most recent month.
- Line chart: Monthly Revenue Trend across all 6 months.
- Combo chart: Monthly Revenue (bars) vs. Margin % (line, secondary axis).
- Add sparklines in the RawData sheet for each store's monthly revenue trend.

Step 4 — Record Macros (Chapters 1 & 2)

- Macro 1 (Absolute): 'FormatRawData' — Selects fixed header range (A1:G1), applies bold, blue fill, white font, and freezes the top row.
- Macro 2 (Relative): 'HighlightNegativeMargin' — Starting from wherever the cursor is, checks the next 24 rows and highlights any margin below 15% in red.
- Macro 3: 'RefreshAll' — Runs FormatRawData, then refreshes all charts. Assign this to a button on the Dashboard.

Step 5 — Add Basic VBA Enhancement (Chapter 3)

Open the VBA Editor (Alt + F11). Find the RefreshAll macro and add a message box at the end:

Step 6 — Protect the Workbook (Chapter 6)

- On the RawData sheet: Unlock the data entry range (A2:G200). Protect the sheet with password 'data123'. Allow selection of unlocked cells only.
- On the Dashboard sheet: Protect the entire sheet. No cells should be editable by users.
- On the Settings sheet: Unlock only the store selection dropdown cell. Protect the sheet.
- Protect the Workbook Structure so sheets cannot be added, deleted, or renamed.

Step 7 — Final Review and Testing

- Enter test data and click the 'Refresh Dashboard' button. Verify the KPI cards and charts update.
- Verify that protected cells cannot be edited. Verify that the unlocked input cells work correctly.
- Change the store selection dropdown on the Settings sheet and verify that charts respond.
- Save the file as .xlsm format. Close and reopen — verify all macros are intact.